

The Higgs Bridge and the Metric Envelope

An AASC Carrier-Order Witness for Hierarchy-Type Comparison

Amos Jay Maley

June 15, 2026

Abstract

This paper constructs an Admissibility And Standing Constraint (AASC) carrier-order witness for Higgs/metric-envelope hierarchy-type comparison. AASC is used here as a necessity-driven constraint system for admissibility, standing, reference, redescription, and witness preservation. The result is not a derivation of the Higgs vacuum expectation value, the Higgs pole mass, Newton’s constant, the Planck mass, or the empirical electroweak/Planck ratio. Its central claim is instead that the Higgs bridge/fixation carrier is standing-subordinate to the metric-envelope carrier in the inherited Higgs/electroweak/gravity comparison scope.

The denominator object is not a Planck label but the metric-envelope carrier, built from first licensed metric comparison, metric-circuit closure, bounded transport, and early transport/closure discipline. The numerator object is not a vacuum expectation value, scalar-potential coordinate, Yukawa term, mass matrix, or empirical weak scale, but the Higgs bridge/fixation carrier inside the realized physical interior. Using standing monotonicity, metric-interior closure, and the realized-physical-interior role ladder, the paper constructs the witness relation

$$C_H \triangleleft C_G^{\text{env}}.$$

The scale-ratio expression is then treated only as a faithful-projection order display. Any faithful positive closure-scale projection places the Higgs bridge carrier below the metric-envelope carrier, yielding

$$0 < \rho_{H/G}^{\text{env}} < 1$$

as proper order type. Projection non-uniqueness prevents this from becoming a derived magnitude, suppression class, quotient-singleton, or empirical $v_F/\bar{M}_{\text{Pl}}$ identification. The paper’s endpoint is therefore carrier order plus faithful-projection hierarchy type, not numerical hierarchy closure.

Contents

Scope and claim boundary		5
1 From no-promotion to positive witness construction		6
1.1 The previous boundary		6
1.2 Why explicit anchor import is lawful but insufficient		6
1.3 The target grammar		6
1.4 Constructive strategy		7

2	Proof-register architecture	7
2.1	Certificate construction and claim-status control	7
2.2	The witness-construction principle	7
2.3	Proof roles of the three recent additions	8
2.4	The exact new contribution	8
2.5	Public-facing interpretation	8
3	Target grammar and ratio modes	8
3.1	Boundary-ratio mode	8
3.2	Candidate structural mode	9
3.3	Derived ratio mode	9
3.4	Closure-scale mode	9
4	Carriers, representatives, and forbidden substitutions	9
4.1	Carrier versus representative	9
4.2	Forbidden substitution rules	10
4.3	Why substitution discipline is necessary	10
5	The denominator: metric-envelope carrier	10
5.1	Definition of the metric-envelope carrier	10
5.2	Metric circuits and no patchwork onset	10
5.3	Closure stress and successor-neighborhood thinness	11
6	The numerator: Higgs bridge/fixation carrier	11
6.1	Definition of C_H	11
6.2	Role-ladder dependence	12
6.3	No vev substitution	12
7	Joint realization of the two carriers	12
7.1	No amplification rule	13
8	Standing monotonicity and carrier order	13
8.1	Standing precedence	13
8.2	Alternative blockers	13
9	Inherited closure-load and faithful scale projection	14
9.1	Why carrier order is not automatically numerical scale	14
9.2	Inherited HEG metric-envelope comparison class	15
9.3	Bridge-anchor diagnostic for the HG target	15
9.4	Unit equivalence and monotone reparameterization	16
9.5	Faithful closure-scale projection	16
9.6	Proper interior response carriers	17
10	The close witness and the faithful-projection order theorem	19
10.1	Definition of the close witness	19
10.2	Well-formedness conditions	19
11	Endpoint registry and status machine	21
11.1	Status table	21

12 Boundary display, empirical ratio, and one-anchor reconstruction	21
12.1 Boundary display	21
12.2 Identification theorems required for empirical readout	22
12.3 One-anchor reconstruction boundary	22
13 Suppression-class test	22
13.1 Why faithful-projection order type is not suppression forcing	22
13.2 Candidate finite residues	23
13.3 What would close Section 11	23
14 Finite-residue search protocol	23
14.1 Candidate finite residues	23
15 Certificate templates	24
15.1 Cross-carrier witness template	24
15.2 Scale-fiber certificate	25
15.3 Endpoint decision procedure	25
16 Worked comparison: boundary ratio versus closure-scale order display	26
16.1 Boundary display comparison	26
16.2 Closure-scale comparison	26
16.3 Review table	26
17 Claim-status safeguards	26
17.1 No hidden empirical ancestry	26
17.2 No naturalness selector	26
17.3 No bridge-to-scale amplification	27
17.4 No Planck-label denominator	27
17.5 No finite-integer shortcut	27
17.6 No one-anchor before shape	27
18 Objections and replies	27
18.1 This still does not derive the empirical weak/Planck ratio	27
18.2 The closure-scale projection is not a physical energy	27
18.3 Carrier order is not numerical order	27
18.4 The projection is an explicit theorem field	27
18.5 Why not use the observed weak and Planck scales directly?	28
18.6 Could future work close a finite suppression class?	28
19 Route exhaustion and counterexample burden	28
20 Theorem ladder and dependency map	29
20.1 Theorem ladder	29
20.2 Dependency map	29
21 Continuation program	29
21.1 Identification maps	29
21.2 Suppression-class route	30
21.3 Quotient-singleton route	30

22 Relation to existing AASC and physics literature	30
22.1 AASC lineage	30
22.2 Physics comparators	30
22.3 Inflation analogy	31
23 Proof architecture	31
24 Order-type closure analysis	31
24.1 Test statement	31
24.2 Alternatives to be obstruction-closed	31
24.3 Faithful-projection order result	32
25 Suppression-class boundary analysis	32
25.1 Test statement	32
25.2 Candidate residues	32
25.3 Suppression-class non-promotion theorem	33
25.4 Current suppression endpoint	33
26 Relation to charged-lepton and neutrino anchor import	33
27 Relation to standard hierarchy language	33
28 Future exactness program	34
28.1 Suppression-class path	34
28.2 Projective-shape path	34
28.3 Quotient-singleton path	34
29 Extension checklist for future uses	34
30 Conclusion	35
A Corpus and source dependency ledger	36
B Certificate checklist	36
C Formalization boundary	37
D Notation index	37
E Extended source import notes	37
E.1 Arrow paper import	37
E.2 Early metric-interior import	37
E.3 RPI import	38
F Publication-facing theorem ladder	38
G Extension protocol	38

Scope and claim boundary

Main claim

This paper constructs a witness, rather than searching for a ready-made one. The witness is designed to close the strongest result currently justified by the available AASC modules: a faithful-projection proper order type for a metric-envelope/Higgs-bridge closure-scale target. The paper does not claim that this ratio is already the observed electroweak/Planck ratio. In the closed endpoint, “ratio” means an order-ratio display under a faithful positive projection; only the order type is invariant. The empirical ratio becomes relevant only as a boundary display or later identification target.

Claim-status ledger

Claim	Status
The metric-envelope carrier C_G^{env} is admitted	Proved/imported and restated
The Higgs bridge carrier $C_H = H_{\text{br}/\text{fix}}$ is admitted	Proved/imported and restated
The cross-carrier witness W_{close}^{HG} is well formed	Main construction
The carrier order $C_H \triangleleft C_G^{\text{env}}$ is forced	Main theorem
The faithful-projection order display satisfies $0 < \rho_{H/G}^{\text{env}} < 1$	Closed order-type display; magnitude not fixed
The empirical ratio $v_F/\bar{M}_{\text{Pl}}$ is derived	Not claimed
A finite suppression class $\rho_{H/G}^{\text{env}} \in \mathcal{S}_k$ is forced	Not claimed; candidate route only
A quotient-singleton value $\rho_{H/G}^{\text{env}} = \rho_*$ is derived	Not claimed
One-anchor reconstruction is lawful	Only after ratio, projective, hierarchy-class, or suppression-class closure

Dependency posture

The paper uses four recent AASC modules as active inputs. The arrow paper supplies quotient-form standing monotonicity, where standing precedence on raw stages becomes a quotient partial order after standing equivalence. The early metric-interior paper supplies the first licensed metric-comparison regime, metric circuits, closure stress as successor-neighborhood thinness, and the early transport/closure certificate. The realized physical interior paper supplies the role ladder from non-nullity through locality, comparison, bounded transport, field carriers, particles, gauge, mass, vacuum, time, and causality, plus the Standard Model structural specialization and the Higgs bridge/fixation carrier. The no-promotion hierarchy paper supplies the HEG redescription discipline, boundary-ratio separation, Planck-label quotient, and false-anchor firewall. The zero-point-energy trilogy supplies quotient/standing discipline for scalar-offset non-promotion in the homogeneous vacuum sector and is cited by DOI [5–7]. This paper has not yet received a paper-specific Lean 4 formalization; the general AASC formalism is mechanized elsewhere in the corpus.

1. From no-promotion to positive witness construction

1.1 The previous boundary

The no-promotion hierarchy analysis established that a scalar representative, cutoff correction, fitted electroweak input, naturalness predicate, threshold split, RG chart value, or Planck label cannot become an internally derived scale anchor merely by being written in a hierarchy comparison. That result is useful but negative. It blocks invalid derivations; it does not construct the positive carrier relation needed for a hierarchy-ratio theorem.

The present paper asks the next question. Once false anchors are excluded, is there a lawful carrier relation between the Higgs bridge and the gravitational or metric-envelope carrier? The answer is yes, provided the target is named correctly. The constructible target is not the observed ratio $v_F/\bar{M}_{\text{Pl}}$. It is the metric-envelope/Higgs-bridge closure-scale ratio:

$$\rho_{H/G}^{\text{env}} = \frac{\sigma_{\text{cl}}(C_H)}{\sigma_{\text{cl}}(C_G^{\text{env}})}. \quad (1)$$

Here σ_{cl} is a closure-scale projection, not an observed energy calibration. The ratio is structural and projected; it becomes empirical only if later identification theorems connect $\sigma_{\text{cl}}(C_H)$ to v_F and $\sigma_{\text{cl}}(C_G^{\text{env}})$ to $\bar{M}_{\text{Pl}}$. At the closed endpoint of this paper, the quotient notation is an order-ratio display. The inequality is invariant under faithful positive projection; the numerical quotient is not invariant without a further normalization, projective-shape, finite-suppression, or scale-fiber-collapse witness.

1.2 Why explicit anchor import is lawful but insufficient

AASC does not forbid anchor import. The charged-lepton and neutrino ratio arcs use lawful one-anchor or boundary-anchor displays after structural ratio, shape, or standing-admission closure. The disciplined pattern is:

$$\text{structural ratio/shape first} \quad \longrightarrow \quad \text{anchor calibration second.}$$

The anchor does not create the ratio. It renders a closed ratio or projective shape in physical units.

For the Higgs/gravity problem, importing both v_F and $\bar{M}_{\text{Pl}}$ gives

$$\rho_{H/G}^{\text{bdry}} = \frac{v_F}{\bar{M}_{\text{Pl}}}. \quad (2)$$

This is meaningful boundary-ratio mode. It is not a derivation. The positive task is to construct the carrier relation that can support a pre-anchor order class. This paper constructs that relation at the level of metric-envelope closure scale.

1.3 The target grammar

There are four distinct targets.

Target	Form	Status
Boundary display	$v_F/\bar{M}_{\text{Pl}}$	physical comparison, not derivation
Candidate ratio	$\sigma_{\text{cl}}(C_H)/\sigma_{\text{cl}}(C_G^{\text{env}})$ before order proof	admissible target
Faithful-projection or- der type	$0 < \rho_{H/G}^{\text{env}} < 1$	closed order display; value projection- dependent
Explicit ratio	$\rho_{H/G}^{\text{env}} = \rho_*$	not claimed
Suppression class	$\rho_{H/G}^{\text{env}} \in \mathcal{S}_k$	candidate route, not forced

1.4 Constructive strategy

The proof proceeds in five stages. First, define the denominator as a metric-envelope carrier rather than a Planck label. Second, define the numerator as a Higgs bridge/fixation carrier rather than a vev or scalar coordinate. Third, use standing monotonicity to prove carrier order. Fourth, use closure stress and metric-envelope maximality to define a closure-scale projection. Fifth, classify the endpoint: faithful-projection proper order type, not projection-unique magnitude, singleton ratio, or finite suppression class.

2. Proof-register architecture

2.1 Certificate construction and claim-status control

The manuscript is organized as a traceable certificate construction. Every object that enters the main theorem is assigned one of four statuses: carrier, projection, boundary display, or blocker. This is essential because the hierarchy problem is especially vulnerable to role drift. A vev can be a fitted electroweak coordinate in one paragraph and a supposed derivation in the next. A Planck mass can be a normalized display of gravitational coupling in one calculation and a primitive denominator in another. The paper prevents that drift by holding the witness fields fixed.

The proof register is deliberately conservative. A theorem reports a fixed-scope necessity or closure result. A proposition reports a classification or dependency consequence. A blocker ledger records why an apparent alternative does not survive as a same-scope alternative. A nonclaim prevents the endpoint from being overreported. This is not merely presentation discipline; it is part of the mathematical posture of AASC. If a proof silently changes the claim type, the proof has changed scope.

2.2 The witness-construction principle

The witness-construction principle used throughout is:

A ratio becomes derivationally meaningful only after its carriers, comparison object, scale projection, transport ledger, calibration ancestry, and blocker alternatives have been declared.

The earlier no-promotion theorem established that scalar coordinates and empirical anchors cannot fill these fields by default. This paper fills the fields explicitly.

2.3 Proof roles of the three recent additions

The three recent additions are not cited as a single opaque authority. Their proof roles are separate.

- Standing monotonicity supplies the order form: raw standing precedence quotiented by standing equivalence yields a partial order.
- Early metric-interior transport closure supplies the denominator: first metric-comparison regime, metric circuit, bounded transport, and early transport/closure certificate.
- The realized physical interior supplies the numerator's dependency position: the Higgs bridge carrier is downstream of comparison, bounded transport, field, particle, gauge, and mass/deformation-response roles.

The theorem requires all three. Without standing monotonicity, carrier order would be informal. Without early metric closure, the denominator would collapse back to a Planck label. Without RPI, the numerator would collapse back to a scalar-coordinate or vev-like representative.

2.4 The exact new contribution

The new contribution is not any one source theorem. It is their composition into the closing witness

$$W_{\text{close}}^{HG} = (S_{\text{HG}}, C_G^{\text{env}}, C_H, \preceq_S, \triangleleft, \text{ClStress}_C, \sigma_{\text{cl}}, \text{Ancestry}, \text{Blockers}, R).$$

The proof of $0 < \rho_{H/G}^{\text{env}} < 1$ is therefore a composite theorem about order display. It rests on carrier admission, bridge-anchor role fixation, carrier-order obstruction closure, and faithful closure-scale projection. The value of the quotient remains projection-dependent until a further normalization or scale-fiber collapse witness is supplied.

2.5 Public-facing interpretation

A reader not using the AASC corpus may read the paper as a foundations result about target adequacy. It says that before asking whether the weak scale is small compared with gravity, one must identify whether the comparison is between measured inputs, formal coordinates, or structural carriers. The paper gives a positive structural-carrier comparison and closes the order type of that comparison.

3. Target grammar and ratio modes

3.1 Boundary-ratio mode

The boundary display is

$$\rho_{H/G}^{\text{bdry}} = \frac{v_F}{\bar{M}_{\text{Pl}}}, \quad v_F = (\sqrt{2} G_F)^{-1/2}, \quad \bar{M}_{\text{Pl}} = (8\pi G_N)^{-1/2}. \quad (3)$$

This ratio is physically meaningful. Both entries are boundary-normalized physical quantities. It is not an AASC derivation because both numerator and denominator are imported from empirical normalization or gravitational measurement.

3.2 Candidate structural mode

The candidate structural ratio is

$$\rho_{H/G}^{\text{cand}} = \frac{\sigma_{\text{cl}}(C_H)}{\sigma_{\text{cl}}(C_G^{\text{env}})}. \quad (4)$$

This is not a number until the carriers, scale map, comparison class, and calibration firewall are declared. It is a target object, not an output.

3.3 Derived ratio mode

An explicit derived ratio would be

$$\rho_{H/G}^{\text{der}} = \rho_*, \quad (5)$$

where ρ_* is invariant under admissible HEG redescription, has no calibration ancestry, and survives transport, normalization, quotient, and obstruction tests. This paper does not reach that endpoint.

3.4 Closure-scale mode

The paper's positive endpoint is the closure-scale hierarchy type

$$0 < \rho_{H/G}^{\text{env}} < 1. \quad (6)$$

This is stronger than target admission and weaker than an explicit numerical ratio. It says that every faithful closure-scale projection displays the Higgs bridge carrier below the metric-envelope carrier in inherited closure-load order; it does not select a unique magnitude.

4. Carriers, representatives, and forbidden substitutions

4.1 Carrier versus representative

A carrier is the standing-bearing role that supports a target. A representative is a notation, coordinate, display, or physical realization of that role inside a chosen scope. The hierarchy problem frequently becomes ambiguous because representatives are promoted into carriers. The current construction forbids that move.

Object	Possible use	Status in this paper
$C_H = H_{\text{br/fix}}$	Higgs bridge/fixation carrier	structural numerator
v_F	electroweak input normalization	boundary display
$v(Q)$	running vev-like coordinate	transport representative
Higgs pole mass	measured excitation response	witness/phenomenological record, not numerator carrier
C_G^{env}	metric-envelope carrier	structural denominator
$\bar{M}_{\text{Pl}}$	gravitational normalization display	boundary display
G_N	measured gravitational coupling	boundary-normalized physical parameter
closure stress	successor-neighborhood thinness	structural order, not stress-energy

4.2 Forbidden substitution rules

The following substitutions are not allowed inside the derivational part of the proof:

$$C_H \not\rightsquigarrow v_F, \quad m_h, \quad \mu^2, \quad V(H), \quad v(Q), \quad (7)$$

$$C_G^{\text{env}} \not\rightsquigarrow \bar{M}_{\text{Pl}}, \quad M_{\text{Pl}}, \quad G_N^{-1/2}, \quad \Lambda_{\text{UV}}, \quad (8)$$

$$\rho_{H/G}^{\text{env}} \not\rightsquigarrow v_F / \bar{M}_{\text{Pl}}. \quad (9)$$

Each substitution may be lawful in display mode, model mode, or empirical comparison mode. It is not lawful in the proof of the structural ratio unless an identification theorem is supplied.

4.3 Why substitution discipline is necessary

Without substitution discipline the theorem would be circular. If the numerator were v_F , the proof would import the observed weak scale. If the denominator were $\bar{M}_{\text{Pl}}$, the proof would import the observed gravitational scale. If the ratio were the empirical quotient, the theorem would merely restate a measured hierarchy. The present proof avoids all three moves.

5. The denominator: metric-envelope carrier

5.1 Definition of the metric-envelope carrier

The denominator is not a Planck label. It is the metric-envelope carrier

$$C_G^{\text{env}} = (L_D, \text{MetricCircuit}_D, \text{BoundTrans}_D, \text{ETC}_D). \quad (10)$$

The entries have the following roles:

- L_D : first licensed metric-comparison regime, a standing class in which quotient-stable comparison is licensed;
- MetricCircuit_D : standing-preserving redescription loop returning metric surrogates to equivalent comparison content;
- BoundTrans_D : bounded perturbation transport discipline;
- ETC_D : early transport/closure certificate relating first metric comparison to later coherent witnesses when local trajectory continuation is insufficient.

The early metric-interior paper is crucial because it treats L_D as a licensing status rather than as a first temporal instant. It also proves that metric comparison requires metric circuits and that early transport/closure is the structural role occupied by inflation-like explanations in metric-field-vacuum projection [2]. Thus the denominator carrier is a metric-interior closure envelope, not a measured Planck mass.

5.2 Metric circuits and no patchwork onset

Licensed metric comparison cannot be a disconnected proto-metric patchwork. If metric comparison is standing-bearing, its comparison content must survive lawful redescription. The early metric-interior theorem therefore yields the implication

$$\text{MetricCmpLicensed}(x) \Rightarrow \text{MetricCircuit}(x). \quad (11)$$

A proto-metric patchwork would require standing transfer across unlicensed patch boundaries, later synchronization repair, a hidden selector, or a scope change. Those routes are not same-scope admissible. This matters for the hierarchy ratio because a Planck-scale denominator cannot be treated as a detached label. The denominator must be the standing-preserving comparison envelope in which physical scale itself has a role.

5.3 Closure stress and successor-neighborhood thinness

The early metric paper defines closure stress by successor-neighborhood inclusion:

$$\text{ClStress}_C(x) \leq \text{ClStress}_C(y) \iff \text{Succ}_C(y) \subseteq \text{Succ}_C(x). \quad (12)$$

Higher closure stress means a thinner admissible successor neighborhood. The first licensed metric-comparison regime is closure-stress maximal relative to its onset-side comparison class. That result is not a claim about physical stress-energy; it is a structural order on admissible continuation load. In metric projection, closure stress may be represented by high-energy, curvature-response, source-load, or vacuum-load language, but those are downstream representatives.

Definition 5.1 (Metric-envelope carrier). A metric-envelope carrier in a fixed HEG scope is a standing-bearing tuple $C_G^{\text{env}} = (L_D, \text{MetricCircuit}_D, \text{BoundTrans}_D, \text{ETC}_D)$ such that L_D licenses quotient-stable comparison, MetricCircuit_D closes comparison under redescription, BoundTrans_D constrains perturbation transport, and ETC_D records early-to-late coherence support where such support is in scope.

Theorem 5.2 (Metric-envelope carrier admission). *The tuple*

$$C_G^{\text{env}} = (L_D, \text{MetricCircuit}_D, \text{BoundTrans}_D, \text{ETC}_D)$$

is an admissible denominator carrier for the Higgs/gravity closure-scale ratio. It supplies the comparison envelope in which scale-bearing physical roles can be projected; it does not by itself supply a numerical Planck value.

Proof. The first metric-comparison regime supplies licensed quotient-stable comparison. Metric circuits ensure that comparison content is not inert display but survives standing-preserving redescription. Bounded transport supplies non-pathless continuation discipline. The early transport/closure certificate supplies the support relation needed when later coherent witnesses impose an early-to-late burden. These four entries jointly carry the metric-envelope role. None of them is a measured Planck label. Therefore C_G^{env} is admitted as denominator carrier, while $\sigma_{\text{cl}}(C_G^{\text{env}})$ still requires declared projection and anchor status. $\square$

6. The numerator: Higgs bridge/fixation carrier

6.1 Definition of C_H

Define

$$C_H := H_{\text{br/fix}}. \quad (13)$$

The realized physical interior paper identifies $H_{\text{br/fix}}$ as the Higgs-like bridge/fixation carrier in the Standard Model structural tuple. It explicitly separates this carrier from a raw scalar coordinate, vev value, potential coefficient, normalization convention, empirical weak scale, Yukawa term, mass matrix, or mixing datum [3]. This separation is essential. The ratio target cannot use the observed weak scale as numerator if it aims at derivational mode.

6.2 Role-ladder dependence

The RPI role ladder places mass and inertia downstream of quotient-stable comparison, local continuation domains, bounded perturbation transport, field carriers, localized persistence classes, and gauge compatibility. Mass is a deformation-response role, not a primitive scalar. The Higgs bridge/fixation carrier therefore enters inside a mature interior architecture:

$$Cmp_D \rightarrow \text{BoundTrans}_D \rightarrow \Phi_D \rightarrow \Pi_D \rightarrow G_D \rightarrow M_D \rightarrow \Omega_D \rightarrow H_{\text{br/fix}}. \quad (14)$$

The arrow here records dependency, not metric time. A Higgs bridge carrier has scale-bearing interpretation only within the comparison and transport envelope that makes deformation response meaningful.

Definition 6.1 (Higgs bridge/fixation carrier). The Higgs bridge/fixation carrier $C_H = H_{\text{br/fix}}$ is the standing carrier mediating gauge/representation order into deformation and mass-response standing over the Standard Model structural skeleton. It is not the empirical vev, not the Higgs pole mass, not a scalar potential coordinate, and not an RG chart value.

Theorem 6.2 (Higgs bridge carrier admission). *The Higgs bridge/fixation object $C_H = H_{\text{br/fix}}$ is an admissible numerator carrier for a Higgs/gravity closure-scale ratio. It supports a candidate scale projection $\sigma_{\text{cl}}(C_H)$, but it does not by itself determine a numerical weak scale.*

Proof. The RPI and Standard Model structural sources admit the Higgs-like bridge/fixation carrier as the structure required to mediate gauge-compatible order into deformation and mass-response standing. The no-scale-from-presentation boundary blocks the substitution of vev, potential coefficient, pole mass, or empirical electroweak input for that carrier. Therefore C_H is an admitted numerator carrier, while its scale projection remains a declared projection object subject to calibration and transport discipline. $\square$

6.3 No vev substitution

Replacing C_H with v_F , a running $v(Q)$, or an effective-potential minimum changes the mode. The result becomes boundary-normalized or presentation-relative unless an additional theorem identifies the representative with $\sigma_{\text{cl}}(C_H)$. This paper does not use that substitution.

7. Joint realization of the two carriers

The carriers must not merely exist separately. They must be jointly realizable inside the same comparison scope.

Definition 7.1 (Joint HEG realization). *JointReal_{HG}(C_H, C_G^{env})* holds when C_H and C_G^{env} are admitted in a single scope S_{HG} , share a quotient comparison object, preserve their carrier identities under redescription, and do not obtain their standing from hidden empirical anchors.

Proposition 7.2 (Joint realization). *JointReal_{HG}(C_H, C_G^{env}) holds for the witness $W_{\text{close}}^{\text{HG}}$.*

Proof. The metric-envelope carrier supplies comparison and transport envelope; the Higgs bridge carrier is admitted as an interior response carrier inside the RPI/SM structural scope. The HEG comparison scope is explicitly fixed to contain both carriers and the closure-scale projection between them. The calibration firewall excludes empirical anchor ancestry. Therefore the carriers are jointly realized as structural carriers rather than as measured values. $\square$

7.1 No amplification rule

Joint realization does not allow amplification. Metric-envelope carrier standing does not derive a Planck value. Higgs bridge standing does not derive a weak-scale value. Their relation gives a carrier-order witness only after the order and projection fields are declared.

Proposition 7.3 (No carrier-to-value amplification). *Neither C_G^{env} nor C_H may be amplified into a numerical scale value without a scale projection and anchor-status declaration.*

Proof. Carrier admission establishes standing of a role. A numerical value requires a scale map, normalization, and either structural collapse or anchor declaration. No such value follows merely from the carrier definitions. Therefore reporting a number from carrier admission would overstate the witness. $\square$

8. Standing monotonicity and carrier order

8.1 Standing precedence

The arrow paper supplies the key monotonicity tool. It defines standing precedence $e_i \preceq_S e_j$ when standing fixed at e_i is retained, transported, presupposed, conditioned upon, or inherited in e_j . Raw stages form a preorder. Quotienting by mutual standing equivalence yields a partial order. The strict quotient direction is non-invertible [1].

For the HEG ratio, the relevant items are not temporal stages but standing-bearing carrier roles inside a fixed comparison scope. We therefore define carrier standing precedence by direct analogy.

Definition 8.1 (HEG carrier standing precedence). For carriers X, Y in S_{HG} , write $X \preceq_S Y$ when the standing-bearing role content of X is retained, transported, presupposed, conditioned upon, or inherited in the admissible use of Y . Write $[X]_S < [Y]_S$ when this precedence is strict after quotienting by carrier-standing equivalence.

Definition 8.2 (Carrier subordination). Write $X \triangleleft Y$ when $Y \preceq_S X$ and the use of X as a scale-bearing interior carrier presupposes Y , while Y does not presuppose X for its carrier standing. Thus $C_H \triangleleft C_G^{\text{env}}$ means that the Higgs bridge carrier is interior/subordinate to the metric-envelope carrier.

8.2 Alternative blockers

The carrier-order proof needs to block equality, reverse order, and incomparability.

Alternative	Collapse mode
$C_H = C_G^{\text{env}}$	identifies Higgs bridge/mass-response carrier with metric-envelope comparison carrier; role collapse
$C_G^{\text{env}} \triangleleft C_H$	makes metric comparison and bounded transport depend on Higgs bridge standing; metric-response backflow
$C_H \parallel C_G^{\text{env}}$	denies common comparison after S_{HG} admits both carriers into one ratio target; target failure
Boundary substitution	replaces carrier relation by empirical anchor comparison; boundary display
Hidden Planck label	imports denominator scale without metric-envelope witness; anchor smuggling
Hidden vev	imports numerator scale without Higgs-bridge scale map; calibration ancestry

Theorem 8.3 (HEG carrier-order witness). *In a fixed HEG scope preserving the AASC kernel, metric-envelope carrier, RPI role ladder, and Higgs bridge/fixation carrier, the Higgs bridge carrier is carrier-subordinate to the metric-envelope carrier:*

$$C_H \triangleleft C_G^{\text{env}}. \quad (15)$$

Equivalently, $C_G^{\text{env}} \preceq_S C_H$ in carrier standing precedence, with strict role distinction after quotienting.

Proof. The metric-envelope carrier supplies licensed comparison through L_D , standing-preserving comparison closure through metric circuits, and bounded transport through BoundTrans_D . The RPI role ladder requires comparison and bounded transport before field, particle, gauge, mass, and vacuum roles can support mass-response standing. The Higgs bridge/fixation carrier is a mass-response bridge inside that mature interior. Therefore the use of C_H as a scale-bearing bridge presupposes the metric-envelope carrier. Conversely, C_G^{env} does not require C_H to license metric comparison, support metric circuits, or constrain bounded transport. Equality collapses distinct roles: metric-envelope comparison is not Higgs bridge mediation. Reverse order would make metric comparison downstream of Higgs bridge standing and so would backflow a mass-response bridge into the conditions that license scale-bearing comparison. Incomparability contradicts the declared HEG ratio target, which admits both carriers into one comparison scope. The only same-scope standing-preserving relation left is $C_H \triangleleft C_G^{\text{env}}$. $\square$

Corollary 8.4 (Carrier-order endpoint). *Before any numerical scale projection is declared, the Higgs/gravity target already has the endpoint*

$$\text{Endpoint}_{HG} \supset \text{MetricEnvelopeWitness} + \text{CarrierOrder} + \text{CandidateRatio}. \quad (16)$$

9. Inherited closure-load and faithful scale projection

9.1 Why carrier order is not automatically numerical scale

Carrier order is not a numerical inequality by itself. Structural precedence, role dependence, and standing inheritance do not always imply larger or smaller physical magnitude. The hierarchy result therefore requires a projection discipline. The paper uses closure stress and successor-neighborhood thinness to define such a discipline only for the declared metric-envelope/Higgs-bridge closure-scale comparison.

The crucial point is that monotonicity is not added as an arbitrary physical postulate. A map that claims to be a closure-scale projection must preserve the strict closure-load order it is supposed to display. If it reverses that order, erases strict order, or reports an independent scale selector, it is not a faithful closure-scale projection of the target. It is a representative, display convention, hidden anchor, or skin.

9.2 Inherited HEG metric-envelope comparison class

Definition 9.1 (Inherited HEG metric-envelope comparison class). The inherited HEG metric-envelope comparison class $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$ consists of the metric-envelope carrier and proper interior response carriers whose scale-bearing standing is evaluated by metric-envelope comparison, bounded transport, and RPI role inheritance. It is not the raw onset-side comparison class of the early metric paper and not a global physical-energy order. It is the HEG comparison class obtained by lifting metric-envelope closure discipline into the mature interior.

Definition 9.2 (Inherited but not global comparison). The inherited class $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$ inherits metric-envelope closure-load order only for carriers whose scale-bearing standing is evaluated through the metric-envelope comparison circuit. It does not compare arbitrary physical energies, arbitrary local stress-energy values, scalar-potential heights, threshold corrections, cutoff terms, or model-specific high-energy representatives. A model may contain large local energy densities inside a declared physical theory without changing the inherited envelope order used here.

Definition 9.3 (Inherited closure-load order). For $X, Y \in \mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$, write

$$X \prec_{\text{cl}} Y \quad (17)$$

when X occupies a strictly lower inherited metric-envelope closure-load class than Y . This order is induced by successor-neighborhood closure stress after transport through the inherited HEG comparison class. It compares closure-load roles, not arbitrary local energy densities, threshold corrections, or model-specific stress-energy representatives.

Proposition 9.4 (Inherited comparison lift). *The Higgs bridge/fixation carrier C_H and the metric-envelope carrier C_G^{env} are comparable in $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$.*

Proof. The metric-envelope carrier supplies the comparison envelope: first metric comparison, metric circuits, bounded transport, and early transport/closure. The Higgs bridge carrier is a mature interior carrier whose scale-bearing use presupposes comparison, bounded transport, field, gauge, and mass/deformation-response roles. Therefore C_H is not compared with C_G^{env} as an element of the raw onset-side class. It is compared through the inherited HEG envelope: the same metric-envelope comparison and transport architecture that makes C_H scale-bearing also places it inside the inherited comparison class. Incomparability would deny the declared HEG ratio target after both carriers have been admitted; hidden anchor substitution would leave the class. Thus the comparison is well formed in $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$. $\square$

9.3 Bridge-anchor diagnostic for the HG target

Definition 9.5 (HG bridge-anchor diagnostic). The HG bridge-anchor diagnostic is the tuple

$$A_{\text{br}}^{HG} = (W_B^{HG}, M_B^{HG}, R_B^{HG}, \rho_B^{HG}, \eta_B^{HG}, \Omega_B^{HG}),$$

where W_B^{HG} is the bridge witness ledger, M_B^{HG} is the mediation map from metric-envelope comparison to Higgs bridge use, R_B^{HG} is the response signature, ρ_B^{HG} is role-rank separation, η_B^{HG} is the closure-scale diagnostic, and Ω_B^{HG} is the obstruction ledger.

Theorem 9.6 (Bridge-anchor role-fixation adequacy). *If the witness ledger, mediation map, response signature, role-rank separation, scale diagnostic, and obstruction ledger of $A_{\text{br}}^{\text{HG}}$ are invariant under admissible HEG redescription, then the bridge-anchor diagnostic is role-fixed for the HG target. If any one of those fields is absent, the diagnostic either collapses to presentation, imports hidden anchor load, fails to distinguish bridge role from scale value, or leaves the target underdetermined.*

Proof. The six fields are exactly the places where the diagnostic can carry standing. The witness ledger fixes what is being witnessed. The mediation map fixes how metric-envelope comparison is inherited by Higgs bridge use. The response signature and role-rank separation distinguish proper interior bridge response from envelope-level comparison. The scale diagnostic fixes the scale-facing report as an order display rather than a scalar value. The obstruction ledger closes equality, reverse-order, incomparability, hidden-anchor, and bridge-to-scale amplification routes. If all six fields are invariant under admissible HEG redescription, the diagnostic has a fixed role. If one is missing, either the target is not identified, the mediation is unlicensed, the bridge is confused with a value, or an alternative route remains open. Thus role fixation follows from field invariance, while magnitude fixation does not. $\square$

9.4 Unit equivalence and monotone reparameterization

Definition 9.7 (Unit equivalence). For positive scale displays, write $x \sim_{\text{unit}} y$ iff $y = \lambda x$ for a positive common unit-normalization factor $\lambda > 0$, unless a richer normalization ledger is explicitly declared. Unit equivalence captures common rescaling or dimensional display choices. It is narrower than arbitrary strictly increasing reparameterization: common unit rescaling preserves ordinary quotients, while a general monotone reparameterization preserves only order.

Remark 9.8 (Two kinds of non-uniqueness). Common unit rescaling preserves ratios. A general strictly increasing reparameterization preserves order but need not preserve ratios. The present witness closes the order type. It does not supply a normalization witness that would privilege one numerical quotient among all faithful order displays.

9.5 Faithful closure-scale projection

Definition 9.9 (Faithful closure-scale projection). A faithful closure-scale projection is a positive map

$$\sigma_{\text{cl}} : \mathcal{O}_{\text{HG}}^{\text{inh}}(C_G^{\text{env}}) \rightarrow \mathbb{R}_{>0} / \sim_{\text{unit}} \quad (18)$$

from inherited carrier classes to positive scale classes modulo common unit normalization, satisfying four conditions:

1. **standing descent:** σ_{cl} is invariant under admissible HEG redescription;
2. **target faithfulness:** σ_{cl} represents inherited closure-load distinctions rather than an independent scale selector;
3. **strict-order preservation:** if $X \prec_{\text{cl}} Y$, then $\sigma_{\text{cl}}(X) < \sigma_{\text{cl}}(Y)$;
4. **calibration firewall:** σ_{cl} is not fixed by importing v_F , G_F , m_H , G_N , M_{Pl} , $\bar{M}_{\text{Pl}}$, or the observed weak/gravity ratio.

Theorem 9.10 (Projection faithfulness forces monotonicity). *Any admitted closure-scale projection for the HEG target must preserve strict inherited closure-load order. If a map τ from $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$ to positive scale classes does not satisfy*

$$X \prec_{\text{cl}} Y \quad \Rightarrow \quad \tau(X) < \tau(Y), \quad (19)$$

then τ is not a faithful closure-scale projection of the HEG closure-load target.

Proof. The target of a closure-scale projection is inherited closure load. A strict relation $X \prec_{\text{cl}} Y$ is standing-bearing content in the inherited metric-envelope comparison class. If $\tau(X) = \tau(Y)$, the map erases a strict standing-bearing closure-load distinction. If $\tau(X) > \tau(Y)$, the map reverses the order it claims to display. If the reversal or erasure is intended to carry physical force, then an independent scale selector has been imported; if it carries no force, the difference is skin. In neither case does τ faithfully project the closure-load order. Therefore strict-order preservation is not an optional assumption but the target-adequacy condition for a closure-scale projection. $\square$

Remark 9.11. This theorem does not assert that every structural order is a scale order. It says only that a map presented as the scale projection of inherited closure load must preserve the inherited closure-load order. A model may reject this projection and retain carrier order alone, but then it cannot report the faithful-projection order display.

Lemma 9.12 (Projection non-uniqueness). *Let σ_{cl} be a faithful closure-scale projection. A common unit rescaling $\lambda\sigma_{\text{cl}}$, with $\lambda > 0$, preserves the displayed quotient. By contrast, if $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ is strictly increasing but not equivalent to a common unit rescaling on the image of σ_{cl} , then $f \circ \sigma_{\text{cl}}$ preserves the same inherited proper order type while generally changing the displayed quotient:*

$$\frac{f(\sigma_{\text{cl}}(C_H))}{f(\sigma_{\text{cl}}(C_G^{\text{env}}))} \neq \frac{\sigma_{\text{cl}}(C_H)}{\sigma_{\text{cl}}(C_G^{\text{env}})}.$$

Therefore the invariant output of the current witness is order type, not numerical magnitude.

Proof. A positive common unit rescaling multiplies numerator and denominator by the same factor and therefore leaves their quotient unchanged. A strictly increasing reparameterization preserves strict inequalities, so $X \prec_{\text{cl}} Y$ implies $f(\sigma_{\text{cl}}(X)) < f(\sigma_{\text{cl}}(Y))$. Thus the faithful-projection order display is unchanged. But ratios are not invariant under arbitrary strictly increasing reparameterization; affine shifts on a positive interval, nonlinear powers, and other monotone maps generally change the quotient unless they reduce to common multiplicative scaling on the relevant image. A unique magnitude would require an additional normalization, projective-shape, finite suppression, or scale-fiber-collapse witness. The present witness supplies none of those. Hence projection faithfulness fixes order type, not magnitude. $\square$

9.6 Proper interior response carriers

Definition 9.13 (Proper interior response carrier). A carrier X is a proper interior response carrier relative to C_G^{env} when X depends on the metric-envelope carrier for scale-bearing interpretation, is not equal to the metric-envelope carrier, and does not add onset-side metric-closure load to the denominator carrier.

Lemma 9.14 (Higgs bridge as proper interior response carrier). *The Higgs bridge/fixation carrier C_H is a proper interior response carrier relative to C_G^{env} .*

Proof. By Theorem 8.3, C_H depends on C_G^{env} for scale-bearing interpretation. By the Higgs bridge/scale separation, C_H is not the metric-envelope carrier and not a Planck label. It is a bridge/fixation carrier mediating gauge/representation order into deformation and mass-response standing. It does not license metric comparison, supply metric circuits, or instantiate the first metric-comparison regime. Therefore it is a proper interior response carrier relative to C_G^{env} . $\square$

Lemma 9.15 (Envelope maximality in the inherited HEG class). *Within $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$, C_G^{env} is the maximal envelope-level comparison carrier. It carries the first metric-comparison regime, metric circuit, bounded transport, and early transport/closure role whose inheritance makes interior scale-bearing comparison possible.*

Proof. By construction, $C_G^{\text{env}} = (L_D, \text{MetricCircuit}_D, \text{BoundTrans}_D, \text{ETC}_D)$. Each component is envelope-level: L_D licenses comparison, MetricCircuit_D preserves comparison under redescription, BoundTrans_D supplies non-pathless transport discipline, and ETC_D supplies early-to-late closure support. A same-scope object with equal or greater envelope-level load would either be this carrier up to equivalence or would introduce a new envelope anchor. Thus C_G^{env} is maximal among envelope-level carriers in the inherited class. $\square$

Lemma 9.16 (Proper interior non-envelope). *The Higgs bridge carrier C_H is not an envelope-level carrier. It inherits metric comparison and bounded transport; it does not license L_D , supply metric circuits, or instantiate the early transport/closure certificate.*

Proof. The RPI role ladder places mass/deformation response downstream of comparison, bounded transport, field carriers, particle classes, and gauge-compatible standing. The Higgs bridge/fixation carrier belongs to that mature interior response layer. If it licensed the first metric-comparison regime or supplied the metric circuit, it would no longer be a Higgs bridge carrier but a metric-envelope carrier. That is a role change, not an interior response relation. $\square$

Lemma 9.17 (No equal inherited closure-load class). *A proper interior response carrier cannot have the same inherited closure-load class as C_G^{env} in $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$. If it does, it has either ceased to be proper interior or has imported envelope load by hidden anchor motion.*

Proof. Equal inherited closure-load class with C_G^{env} would make the interior carrier perform the same envelope-level work as the object that licenses comparison and transport inheritance. If this work is declared, the carrier has been promoted to envelope status and the theorem object changes. If it is undeclared, it is hidden load. If it changes no standing-bearing content, equality is only skin and does not identify the carrier roles. Therefore no proper interior response carrier shares the envelope closure-load class. $\square$

Theorem 9.18 (Strict inherited closure-load separation). *In $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$, the Higgs bridge carrier occupies a strictly lower inherited closure-load class than the metric-envelope carrier:*

$$C_H \prec_{\text{cl}} C_G^{\text{env}}. \quad (20)$$

Proof. By Proposition 9.4, the carriers are compared only in the inherited HEG metric-envelope comparison class, not in a global energy or stress-energy order. By Lemma 9.15, C_G^{env} is the maximal envelope-level carrier in that class. By Lemma 9.16, C_H is not an envelope-level carrier; it inherits the comparison and transport structure. By Lemma 9.17, C_H cannot occupy the same inherited closure-load class as C_G^{env} . A greater inherited closure-load class for C_H would introduce a new envelope anchor or reverse the RPI dependency order by making metric comparison downstream of Higgs bridge standing. Incomparability is excluded by the inherited comparison lift. Thus the only same-scope admissible relation is $C_H \prec_{\text{cl}} C_G^{\text{env}}$. $\square$

Theorem 9.19 (Faithful closure-scale order display). *For any faithful HEG closure-scale projection σ_{cl} , strict inherited closure-load separation induces strict positive scale separation:*

$$X \prec_{\text{cl}} Y \quad \Rightarrow \quad 0 < \sigma_{\text{cl}}(X) < \sigma_{\text{cl}}(Y). \quad (21)$$

In particular,

$$0 < \sigma_{\text{cl}}(C_H) < \sigma_{\text{cl}}(C_G^{\text{env}}). \quad (22)$$

Proof. Positivity is part of scale-projection status for standing-bearing closure-scale carriers. By Theorem 9.10, a faithful closure-scale projection preserves strict inherited closure-load order. Applying Theorem 9.18 with $X = C_H$ and $Y = C_G^{\text{env}}$ yields $0 < \sigma_{\text{cl}}(C_H) < \sigma_{\text{cl}}(C_G^{\text{env}})$. $\square$

10. The close witness and the faithful-projection order theorem

10.1 Definition of the close witness

Definition 10.1 (Higgs/gravity close witness). The Higgs/gravity close witness is

$$W_{\text{close}}^{HG} = (S_{\text{HG}}, C_G^{\text{env}}, C_H, \preceq_S, \triangleleft, \text{ClStress}_C, \sigma_{\text{cl}}, \text{Ancestry}, \text{Blockers}, R), \quad (23)$$

where S_{HG} is the fixed comparison scope, C_G^{env} is the metric-envelope carrier, C_H is the Higgs bridge carrier, $\preceq_S$ is standing precedence, $\triangleleft$ is carrier subordination, ClStress_C is closure-stress order, σ_{cl} is the closure-scale projection, Ancestry is the calibration firewall, Blockers is the alternative-exhaustion ledger, and R is the endpoint report.

10.2 Well-formedness conditions

The witness is well formed iff the following fields close:

1. C_G^{env} is admitted by Theorem 5.2;
2. C_H is admitted by Theorem 6.2;
3. $C_H \triangleleft C_G^{\text{env}}$ is proved by Theorem 8.3;
4. ClStress_C is inherited from successor-neighborhood thinness;
5. σ_{cl} is a faithful closure-scale projection and A_{br}^{HG} fixes bridge-anchor role status;
6. the ancestry firewall excludes empirical weak-scale and Planck-scale imports;
7. equality, reverse order, incomparability, hidden vev, hidden Planck label, and naturalness selector routes are obstruction-closed;
8. the endpoint report does not exceed the strongest closed status.

Theorem 10.2 (Close-witness construction). *The tuple W_{close}^{HG} is a well-formed AASC cross-carrier witness for the metric-envelope/Higgs-bridge closure-scale hierarchy target.*

Proof. The denominator and numerator carriers are admitted by Theorems 5.2 and 6.2. Standing precedence and carrier subordination are supplied by Theorem 8.3. Closure stress is a structural order over successor neighborhoods. The bridge-anchor diagnostic fixes witness, mediation, response, role-rank, scale-diagnostic, and obstruction fields by Theorem 9.6. The closure-scale projection

is declared with standing descent, target faithfulness, strict-order preservation, and calibration firewall. The blocker ledger excludes equality, reverse order, incomparability, empirical anchor substitution, and hidden selector routes. The endpoint report is bounded to bridge-anchor role fixation, faithful-projection proper order type, projection non-uniqueness, and suppression-candidate status. Thus each field of W_{close}^{HG} has declared role, standing, and boundary status. $\square$

Theorem 10.3 (HEG metric-envelope faithful-projection order theorem). *Given the close witness W_{close}^{HG} , the metric-envelope/Higgs-bridge closure-scale ratio satisfies*

$$0 < \rho_{H/G}^{\text{env}} = \frac{\sigma_{\text{cl}}(C_H)}{\sigma_{\text{cl}}(C_G^{\text{env}})} < 1. \quad (24)$$

Proof. By Theorem 8.3, $C_H \triangleleft C_G^{\text{env}}$. By Lemma 9.14, C_H is a proper interior response carrier relative to C_G^{env} . By Theorem 9.18, the inherited closure-load separation is strict:

$$C_H \prec_{\text{cl}} C_G^{\text{env}}. \quad (25)$$

By Theorem 9.19, every faithful closure-scale projection maps strict inherited closure-load separation to strict positive scale separation:

$$0 < \sigma_{\text{cl}}(C_H) < \sigma_{\text{cl}}(C_G^{\text{env}}). \quad (26)$$

Since $\sigma_{\text{cl}}(C_G^{\text{env}}) > 0$, division gives

$$0 < \frac{\sigma_{\text{cl}}(C_H)}{\sigma_{\text{cl}}(C_G^{\text{env}})} < 1. \quad (27)$$

This is exactly the faithful-projection order display $0 < \rho_{H/G}^{\text{env}} < 1$. By Theorem 10.4, the displayed quotient value is not an invariant magnitude unless a further normalization or scale-fiber collapse witness is supplied. $\square$

Theorem 10.4 (No magnitude closure from the current witness). *The witness W_{close}^{HG} does not determine a projection-unique magnitude for $\rho_{H/G}^{\text{env}}$. It determines the faithful-projection proper order type*

$$0 < \rho_{H/G}^{\text{env}} < 1$$

for every faithful positive order projection, while leaving the numerical value of the quotient open.

Proof. The carrier-order and inherited closure-load theorems fix strict order. The bridge-anchor diagnostic fixes role and order-report status. The faithful projection theorem says admissible closure-scale projections preserve that strict order. By Lemma 9.12, strictly increasing reparameterizations preserve the order while generally changing the quotient value. No finite residue controlling σ_{cl} , projective shape, one-anchor scale-fiber collapse, or quotient-singleton witness is proved in this manuscript. Therefore the magnitude is not closed. $\square$

Corollary 10.5 (Endpoint classification). *The current closed endpoint is*

$$\begin{aligned} \text{Endpoint}_{HG} = & \text{MetricEnvelopeWitness} + \text{CarrierOrder} \\ & + \text{BridgeAnchorRoleFixed} + \text{FaithfulProjectionProperOrderType} \\ & + \text{ProjectionNonUniqueness} + \text{SuppressionCandidate}. \end{aligned} \quad (28)$$

It is not $\rho_{H/G}^{\text{env}} = \rho_$, not $\rho_{H/G}^{\text{env}} \in \mathcal{S}_k$ as forced class, and not a derivation of $v_F/\bar{M}_{\text{Pl}}$.*

11. Endpoint registry and status machine

The witness is best understood as a status machine. It accepts declared carriers and projection data and outputs the strongest admissible endpoint.

Definition 11.1 (Endpoint registry). The endpoint registry is

$$Reg_{HG}(W_{close}^{HG}) = (Status, Witnesses, Blockers, NextCert).$$

The admissible statuses are boundary display, candidate ratio, carrier order, faithful-projection proper order type, suppression candidate, suppression forced, projective class, and quotient singleton.

Theorem 11.2 (Endpoint registry theorem). *For the witness W_{close}^{HG} , the registry output is*

$$Reg_{HG}(W_{close}^{HG}) = \text{BridgeAnchorRoleFixed} + \text{FaithfulProjectionProperOrderType} \\ + \text{ProjectionNonUniqueness} + \text{SuppressionCandidate},$$

where *FaithfulProjectionProperOrderType* refers to faithful-projection proper order display $0 < \rho_{H/G}^{\text{env}} < 1$, *ProjectionNonUniqueness* records that the quotient value is not invariant, and *SuppressionCandidate* records open finite-residue routes.

Proof. Carrier admission and joint realization give more than candidate ratio. The carrier-order witness, bridge-anchor diagnostic, strict inherited closure-load separation, and faithful projection theorem give the faithful-projection proper order display. Projection non-uniqueness blocks magnitude closure. No finite residue controlling the scale map is proved, so suppression forcing is not available. No exact quotient-singleton value is proved. Therefore the registry reports bridge-anchor role fixation, proper order type, projection non-uniqueness, and suppression candidate. $\square$

11.1 Status table

Status	Required witness	Current result
Boundary display	empirical anchors	available, nonderivational
Candidate ratio	admitted carriers	available
Carrier order	standing precedence + blockers	closed
Faithful-projection proper order type	bridge-anchor diagnostic + faithful order projection	closed
Projection non-uniqueness	strictly increasing reparameterizations preserve order but alter quotient	closed; magnitude blocked
Magnitude ratio	unique normalization or scale-fiber collapse	not closed
Suppression class	finite residue controlling σ_{cl}	not closed
Quotient singleton	scale-fiber collapse	not closed
Empirical identification	map to $v_F, \bar{M}_{Pl}$	not closed

12. Boundary display, empirical ratio, and one-anchor reconstruction

12.1 Boundary display

The empirical weak/gravity display remains

$$\rho_{H/G}^{\text{bdry}} = \frac{v_F}{\bar{M}_{Pl}}. \quad (29)$$

The close witness does not use this ratio. It may be used after the fact as boundary comparison.

12.2 Identification theorems required for empirical readout

To identify the closure-scale hierarchy with the observed electroweak/Planck hierarchy, two additional maps must be certified:

$$\sigma_{\text{cl}}(C_H) \mapsto v_F, \quad (30)$$

$$\sigma_{\text{cl}}(C_G^{\text{env}}) \mapsto \bar{M}_{\text{Pl}}. \quad (31)$$

Each map requires a witness ledger. The first must show that the closure-scale projection of the Higgs bridge carrier is the electroweak input scale rather than a boundary-normalized representative. The second must show that the metric-envelope closure-scale projection is the reduced Planck scale rather than a conventionally normalized gravitational anchor.

12.3 One-anchor reconstruction boundary

If a stronger theorem later closes a projective shape or explicit ratio,

$$[\sigma_{\text{cl}}(C_H) : \sigma_{\text{cl}}(C_G^{\text{env}})] = [a : b], \quad (32)$$

then one-anchor reconstruction is lawful:

$$\sigma_{\text{cl}}(C_H) = \frac{a}{b} \sigma_{\text{cl}}(C_G^{\text{env}}). \quad (33)$$

But the anchor does not create the shape. Without pre-anchor closure, one-anchor import collapses to boundary display.

Proposition 12.1 (Boundary display separation). *The faithful-projection order theorem for $\rho_{H/G}^{\text{env}}$ does not identify $\rho_{H/G}^{\text{env}}$ with $\rho_{H/G}^{\text{bdry}}$. The empirical ratio is a lawful physical comparison, while $\rho_{H/G}^{\text{env}}$ is a closure-scale hierarchy target.*

Proof. The empirical ratio uses observed or boundary-normalized anchors. The closure-scale ratio uses admitted carriers and a closure-scale projection protected by a calibration firewall. Since the derivational ancestry differs, the ratios cannot be identified without an additional witness mapping the closure-scale projection to the empirical anchors. Therefore the empirical ratio is preserved as boundary display but not promoted to derivational output. $\square$

13. Suppression-class test

13.1 Why faithful-projection order type is not suppression forcing

The proof of $0 < \rho_{H/G}^{\text{env}} < 1$ establishes faithful-projection proper order type. It does not provide a finite exponent, obstruction index, quotient singleton, or suppression class. A suppression theorem would require

$$\rho_{H/G}^{\text{env}} \in \mathcal{S}_k \quad (34)$$

for a standing-bearing finite class $\mathcal{S}_k$. The current witness does not contain such a class.

13.2 Candidate finite residues

The uploaded and matrix materials suggest several candidate residue families:

Candidate	Current status
Family number $N_g = 3$	structural cardinality; not automatically a scale exponent
Charge lattice / hypercharge singleton	charge structure; not scale-map controller
Anomaly cancellation residue	consistency requirement; not magnitude relation
Higgs bridge role rank	plausible role residue; no scale-control theorem yet
Vacuum compatibility obstruction index	plausible candidate; no exponent map yet
Closure-stress codimension	strongest candidate for future work; requires finite codimension theorem
Successor-neighborhood rank	candidate for suppression classification; not currently finite-forced
Metric-circuit minimality index	possible; needs relation to σ_{cl}

Theorem 13.1 (Suppression-class non-closure). *The close witness W_{close}^{HG} does not force a finite suppression class for $\rho_{H/G}^{env}$. It registers a suppression candidate only.*

Proof. The faithful-projection order theorem uses carrier order, closure-stress separation, and faithful closure-scale projection. None of these fields supplies a finite exponent, finite obstruction index, projective shape, or quotient-singleton value. The candidate residues listed above are structural, but no theorem in this manuscript proves that any of them controls the closure-scale map. Promoting them to exponents would violate the no finite-integer shortcut. Therefore the endpoint includes SuppressionCandidate, not forced suppression. $\square$

13.3 What would close Section 11

A future suppression theorem would need a finite-residue control statement:

$$r_{HG} \text{ standing-bearing} \wedge r_{HG} \text{ controls } \sigma_{cl} \Rightarrow \rho_{H/G}^{env} \in \mathcal{S}_{r_{HG}}. \quad (35)$$

The most promising route is a closure-stress codimension or successor-neighborhood rank theorem, because those quantities touch the metric-envelope closure structure directly. Family number and charge residue are less promising unless a separate theorem links them to the closure-scale map.

14. Finite-residue search protocol

The suppression-class test remains open, but the paper can state the protocol for future closure.

14.1 Candidate finite residues

A finite residue can be tested only if it is standing-bearing and if it controls σ_{cl} . Plausible candidate sources include:

1. closure-stress codimension in the onset-side comparison class;

2. successor-neighborhood rank or deficiency;
3. metric-circuit minimality index;
4. early transport/closure obstruction index;
5. Higgs bridge role rank;
6. vacuum-compatibility obstruction count;
7. anomaly-compatible survivor residue;
8. charge-lattice rigidity residue;
9. family-cardinality envelope data.

The last three are especially dangerous: structural integers do not become exponents by coexistence. They become exponents only if a theorem shows that they control the closure-scale map.

Readiness condition 14.1 (Suppression-class readiness). A finite residue r is suppression-ready only if:

1. r has standing independently of empirical anchors;
2. r is invariant under HEG admissible redescription;
3. changing r changes σ_{cl} rather than notation;
4. non- r -controlled assignments are obstruction-closed;
5. the resulting class is transport-stable and calibration-prior.

Proposition 14.2 (No finite-integer shortcut). *Family number, charge-lattice residue, anomaly cancellation, or metric-circuit count does not force a suppression class unless it is proved to control the closure-scale projection.*

Proof. A structural integer may type a carrier, representation, or obstruction. A suppression class requires control of a scale map. Without a theorem linking the integer to σ_{cl} , using it as an exponent is hidden selector import. Therefore no integer shortcut is admitted. $\square$

15. Certificate templates

15.1 Cross-carrier witness template

The general cross-carrier witness can be written

$$W_{\text{rel}}^{HG} = (C_H, C_G^{\text{env}}, Q_{HG}, \sigma_{\text{cl}}, T_{HG}, A_{HG}, N_{HG}, B_{HG}, R_{HG}). \quad (36)$$

The close witness is the specialization where Q_{HG} is the closure-scale ratio, T_{HG} is inherited through metric-envelope bounded transport, A_{HG} is the calibration firewall, and B_{HG} closes equality, reverse-order, incomparability, and anchor-smuggling alternatives.

15.2 Scale-fiber certificate

The scale-fiber certificate is

$$HBSCert_{HG} = (\textit{Target}, \textit{Hierarchy}, \textit{Bridge}, \textit{ScaleFib}, \textit{Anchor}, \textit{Norm}, \textit{Ancestry}, \textit{Collapse}, \textit{Blockers}, \textit{Report}). \quad (37)$$

For this paper:

Field	Status
Target	$\rho_{H/G}^{\text{env}}$
Hierarchy	faithful-projection proper order type $0 < \rho_{H/G}^{\text{env}} < 1$
Bridge	$C_H = H_{\text{br/fix}}$
Scale fiber	not singleton-collapsed
Anchor	none in derivation; empirical display allowed separately
Norm	closure-scale projection, not MeV/Planck display
Ancestry	no $v_F, G_F, m_H, G_N, M_{\text{Pl}}$ input
Collapse	order type only
Blockers	equality, reverse-order, incomparability, hidden anchors
Report	MetricEnvelopeWitness + CarrierOrder + BridgeAnchorRoleFixed + FaithfulProjectionProperOrderType + ProjectionNonUniqueness + SuppressionCandidate

15.3 Endpoint decision procedure

Definition 15.1 (Endpoint classifier). The endpoint classifier returns the strongest status closed by the witness:

$$\text{Endpoint}_{HG} \in \{\text{BoundaryDisplay}, \text{CandidateRatio}, \text{CarrierOrder}, \text{FaithfulProjectionProperOrderType}, \text{SuppressionCandidate}, \text{SuppressionForced}, \text{ProjectiveClass}, \text{QuotientSingleton}\}. \quad (38)$$

Theorem 15.2 (Endpoint decision). For W_{close}^{HG} , the endpoint classifier returns

$$\begin{aligned} \text{Endpoint}_{HG} = & \text{MetricEnvelopeWitness} + \text{CarrierOrder} \\ & + \text{BridgeAnchorRoleFixed} + \text{FaithfulProjectionProperOrderType} \\ & + \text{ProjectionNonUniqueness} + \text{SuppressionCandidate}. \end{aligned} \quad (39)$$

Proof. Carrier admission, witness construction, carrier order, strict inherited closure-load separation and faithful closure-scale projection close the faithful-projection proper order type. The blocker ledger prevents promotion to empirical boundary ratio, quotient singleton, or finite suppression class. The finite-residue candidates remain unproved as scale-map controllers. Therefore the stated endpoint is the strongest supported report. $\square$

16. Worked comparison: boundary ratio versus closure-scale order display

16.1 Boundary display comparison

The boundary display uses two anchors:

$$\rho_{H/G}^{\text{bdry}} = \frac{v_F}{\bar{M}_{\text{Pl}}}. \quad (40)$$

It is a meaningful physical comparison. Its ancestry includes empirical electroweak and gravitational normalization. It cannot be reported as derivational closure.

16.2 Closure-scale comparison

The closure-scale ratio uses:

$$\rho_{H/G}^{\text{env}} = \frac{\sigma_{\text{cl}}(C_H)}{\sigma_{\text{cl}}(C_G^{\text{env}})}. \quad (41)$$

Its ancestry is carrier-theoretic:

$$\text{standing monotonicity} + \text{metric-envelope closure} + \text{RPI role ladder} + \text{Higgs bridge carrier}. \quad (42)$$

It does not import v_F or $\bar{M}_{\text{Pl}}$.

16.3 Review table

Mode	Inputs	Output
Boundary display	$v_F, \bar{M}_{\text{Pl}}$	empirical ratio, no derivation
Candidate target	$C_H, C_G^{\text{env}}, \sigma_{\text{cl}}$ before order proof	admitted target
Carrier order	$C_G^{\text{env}} \preceq_S C_H$	nontrivial witness relation
Closure-scale hierarchy	$C_H \prec_{\text{cl}} C_G^{\text{env}}, \text{faithful } \sigma_{\text{cl}}$	$0 < \rho_{H/G}^{\text{env}} < 1$
Suppression class	finite residue controlling σ_{cl}	not closed
Singleton	total scale-fiber collapse	not closed

17. Claim-status safeguards

17.1 No hidden empirical ancestry

The proof of the faithful-projection order theorem does not use $v_F, G_F, m_H, M_W, M_Z, G_N, M_{\text{Pl}}, \bar{M}_{\text{Pl}}$, or the observed ratio. Any proof that does use them must be marked boundary display or mixed-mode reconstruction.

17.2 No naturalness selector

Naturalness may motivate the search for a scale witness. It does not supply the witness. No step of the proof uses order-one expectation, absence of tuning, radiative-stability preference, or anthropic range as a selector.

17.3 No bridge-to-scale amplification

Higgs bridge standing does not imply a numerical weak scale. The proof uses bridge standing only to admit C_H as a numerator carrier and then uses closure-scale projection to establish faithful-projection proper order type. It does not derive a Higgs vev.

17.4 No Planck-label denominator

The denominator is C_G^{env} , not M_{Pl} . A Planck label can display a gravitational scale after normalization, but it is not the denominator carrier used in the proof.

17.5 No finite-integer shortcut

Family number, charge-lattice structure, anomaly cancellation, and bridge-rank residues are not suppression exponents unless a theorem proves that they control the closure-scale projection.

17.6 No one-anchor before shape

One-anchor reconstruction is allowed only after a ratio, projective shape, hierarchy class, or suppression class is certified. The theorem certifies a faithful-projection order class for $\rho_{H/G}^{\text{env}}$, not an exact shape or singleton.

18. Objections and replies

18.1 This still does not derive the empirical weak/Planck ratio

Correct. It derives a faithful-projection proper order type for the closure-scale projection of the Higgs bridge carrier relative to the metric-envelope carrier. The empirical ratio remains boundary display unless additional identification theorems close.

18.2 The closure-scale projection is not a physical energy

It is not asserted to be a measured energy. It is a standing-preserving scale projection over closure load. Physical energy, curvature-response, stress-energy, or Planck-language displays are downstream representatives.

18.3 Carrier order is not numerical order

Correct in general. This paper does not rely on generic structural precedence. It proves an inherited closure-load separation and restricts the scale step to faithful closure-scale projections of that order. A projection that reverses or erases the order is a different target, not a competing closure-scale map.

18.4 The projection is an explicit theorem field

The projection is not hidden. It is one of the explicit fields of W_{close}^{HG} , and the manuscript proves that a map that fails to preserve inherited closure-load order is not a faithful closure-scale projection of this target. Refusing faithful projection leaves the endpoint at carrier order and candidate ratio, because the scale-projection target has been abandoned. Retaining faithful projection closes

faithful-projection proper order type. The paper therefore states both the carrier-order fallback and the projected hierarchy endpoint transparently.

18.5 Why not use the observed weak and Planck scales directly?

Because that would give only a boundary display. It would not explain why the scale relation is structurally present before calibration. The current proof separates the structural order class from empirical rendering.

18.6 Could future work close a finite suppression class?

Yes. The best candidate route is a finite closure-stress codimension, successor-neighborhood rank, or metric-circuit minimality index controlling σ_{cl} . This paper does not claim that closure.

19. Route exhaustion and counterexample burden

A same-scope rival must preserve the declared target: admitted metric-envelope carrier, admitted Higgs bridge carrier, lawful redescription, no hidden empirical anchors, and the claim that the closure-scale order can be displayed without W_{close}^{HG} . The rival is resolved by the following normal forms.

Route	Expression in this target	Status
Carrier-equivalent witness	preserves C_G^{env} , C_H , order, projection, firewall, and blockers	same witness under redescription
Boundary display	uses v_F , G_F , m_H , G_N , M_{Pl} , or $\bar{M}_{Pl}$	empirical or mixed mode
Richer-scope mechanism	adds new dynamics, fields, cosmological boundary data, or quantum-gravity structure	legitimate only with projection back to the target
Hidden anchor/tensor	does the work without declaring the load	inadmissible
Bridge-to-scale amplification	converts bridge standing into a value without scale map	blocked
Metric backflow	makes the metric-envelope carrier depend on Higgs bridge standing	blocked
Projection refusal	rejects faithful σ_{cl}	leaves carrier order only
Suppression overclaim	reports exponent without finite controlling residue	blocked
Quotient motion	changes carrier or witness identity	target change

A responsive counterexample must either break carrier admission, break the RPI role order, identify deletion-tested roles without collapse, reverse metric-envelope dependency, deny the HEG comparison scope, or reject the faithful closure-scale projection while retaining the same projection target. Importing empirical scales or adding new dynamics changes the mode of explanation rather than refuting the same-scope witness.

20. Theorem ladder and dependency map

20.1 Theorem ladder

The proof ladder can be read as follows.

1. **Metric-envelope admission.** C_G^{env} is the denominator carrier, not a Planck label.
2. **Higgs bridge admission.** $C_H = H_{\text{br/fix}}$ is the numerator carrier, not a vev or pole mass.
3. **Standing precedence import.** Standing monotonicity supplies a quotient-order pattern for carrier precedence.
4. **Carrier-order witness.** $C_H \triangleleft C_G^{\text{env}}$ is forced after equality, reverse order, and incomparability are blocked.
5. **Closure-stress separation.** C_H has lower inherited closure stress than C_G^{env} inside the HEG closure-stress comparison class.
6. **Closure-scale projection.** σ_{cl} maps strict inherited closure-load separation to strict positive scale separation.
7. **Faithful-projection order display.** $0 < \rho_{H/G}^{\text{env}} < 1$ as order type, not magnitude.
8. **Endpoint discipline.** The result is not a singleton, not a finite suppression class, and not the empirical ratio.

20.2 Dependency map

Step	Dependency	Output
Metric envelope	early metric closure, metric circuit, bounded transport, ETC	C_G^{env}
Higgs carrier	RPI mass-response ladder and SM bridge/fixation carrier	C_H
Monotonicity	standing precedence quotient order	carrier order relation
Carrier blockers	role separation and no-backflow	$C_H \triangleleft C_G^{\text{env}}$
Stress separation	closure-stress maximality and proper interior carrier status	$C_H \prec_{\text{cl}} C_G^{\text{env}}$
Scale projection	faithful closure-scale projection over inherited closure load	$0 < \sigma_{\text{cl}}(C_H) < \sigma_{\text{cl}}(C_G^{\text{env}})$
Endpoint classifier	scale-fiber and blocker ledger	order type; magnitude open

21. Continuation program

21.1 Identification maps

The faithful-projection order theorem leaves open two identification maps:

$$I_H : \sigma_{\text{cl}}(C_H) \rightarrow v_F, \quad I_G : \sigma_{\text{cl}}(C_G^{\text{env}}) \rightarrow \bar{M}_{\text{Pl}}. \quad (43)$$

If both maps are later certified without hidden calibration ancestry, then the closure-scale hierarchy can be connected to the empirical electroweak/Planck hierarchy. If only one map is certified, the

result becomes mixed-mode reconstruction. If neither is certified, the present theorem remains a structural hierarchy theorem.

21.2 Suppression-class route

The most promising suppression-class route is a finite closure-stress residue:

$$r_{\text{cl}} = \text{codim}_{HG}(\text{Succ}_C(C_H), \text{Succ}_C(C_G^{\text{env}})) \quad (44)$$

or a related metric-circuit minimality index. Such an index would matter only if it is standing-bearing and controls σ_{cl} . A future theorem would have to prove all of the following:

1. r_{cl} is finite and quotient-stable;
2. r_{cl} is not a presentation artifact;
3. r_{cl} controls the closure-scale map rather than merely coexisting with it;
4. all non- r_{cl} -controlled scale assignments are obstruction-closed;
5. no empirical weak-scale or Planck-scale value appears in the ancestry graph.

Only then would the endpoint advance from `SuppressionCandidate` to `SuppressionForced`.

21.3 Quotient-singleton route

A quotient-singleton result would require total scale-fiber collapse:

$$\rho_{H/G}^{\text{env}} = \rho_*. \quad (45)$$

The present witness does not contain enough information for this. It would require a unique closure-scale map, obstruction exhaustion, normalization closure, transport closure, and no remaining representative freedom. The paper therefore does not gesture at a numerical value.

22. Relation to existing AASC and physics literature

22.1 AASC lineage

The paper relies on three recent AASC developments. Standing monotonicity supplies the quotient-order machinery for non-invertible standing precedence [1]. Early metric-interior transport closure supplies first metric comparison, metric circuits, successor-neighborhood closure stress, and early transport/closure certificates [2]. The realized physical interior supplies the local-domain, comparison, bounded-transport, field, particle, gauge, mass, vacuum, time, causality, gravity, QFT, and Standard Model role stack [3]. The ZPE trilogy supplies the renormalization/standing quotient model for scalar non-promotion [5–7].

22.2 Physics comparators

The hierarchy problem remains tied to the literature on naturalness, scalar sensitivity, effective field theory, and UV completion [8–10, 12–14]. The present result is not an alternative to model building. It is a target-adequacy theorem: a model that aims to explain the weak/gravity hierarchy must eventually identify the carriers and scale maps whose ratio it is explaining.

22.3 Inflation analogy

The early metric-interior paper treats inflation-like roles as early transport/closure certificates, not as primitive causes of metricity. This is analogous to the present paper's treatment of the metric-envelope carrier: the denominator is a structural carrier of metric comparison and closure, not a Planck label imported before standing. In both cases, physical model symbols are admitted as downstream projections, not foundations.

23. Proof architecture

The proof spine is compact. First, standing monotonicity supplies the quotient-order form. Second, the metric-envelope carrier C_G^{env} supplies the denominator without Planck-label import. Third, the Higgs bridge carrier C_H supplies the numerator without vev substitution. Fourth, the inherited comparison lift places both carriers in $\mathcal{O}_{HG}^{\text{inh}}(C_G^{\text{env}})$. Fifth, bridge-anchor role fixation and strict inherited closure-load separation yield $C_H \prec_{\text{cl}} C_G^{\text{env}}$. Sixth, faithful projection gives the order display $0 < \rho_{H/G}^{\text{env}} < 1$, while projection non-uniqueness blocks magnitude overclaim.

Stage	Output
Carrier admission	C_G^{env} and C_H are admitted carriers
Carrier order	$C_H \triangleleft C_G^{\text{env}}$
Inherited closure-load order	$C_H \prec_{\text{cl}} C_G^{\text{env}}$
Faithful projection	$0 < \rho_{H/G}^{\text{env}} < 1$ as order display
Non-uniqueness	no projection-unique magnitude
Endpoint	order-type witness, suppression candidate, magnitude open

24. Order-type closure analysis

24.1 Test statement

The proper-hierarchy test asks whether the HEG target can advance from candidate ratio to an ordered hierarchy type. In the notation of the witness,

$$\text{CandidateRatio} \longrightarrow \text{FaithfulProjectionProperOrderType}$$

requires either a direct scale inequality or a carrier-order theorem plus a legitimate scale projection. The present manuscript supplies the second route.

24.2 Alternatives to be obstruction-closed

The alternatives are:

- (a) equality: $C_H = C_G^{\text{env}}$;
- (b) reverse order: $C_G^{\text{env}} \triangleleft C_H$;
- (c) incomparability: $C_H \parallel C_G^{\text{env}}$;
- (d) boundary display substituted for structural ratio;

- (e) hidden empirical anchor import;
- (f) carrier shift by changing the theorem object.

The carrier-order theorem closes the first three alternatives. The calibration firewall closes hidden empirical anchor import. The target grammar separates boundary display from derivational structure. Scope declaration handles carrier shift.

24.3 Faithful-projection order result

The order-type endpoint is therefore:

$$Status_9 = \text{CarrierOrder} + \text{FaithfulProjection} + \text{FaithfulProjectionProperOrderType}.$$

The faithful projection closes the order report as:

$$Status_9 = \text{FaithfulProjectionProperOrderType}$$

for the metric-envelope closure-scale ratio.

25. Suppression-class boundary analysis

25.1 Test statement

The suppression test asks whether a finite residue, obstruction index, role-rank count, or quotient-collapsed finite carrier controls the ratio strongly enough to force

$$\rho_{H/G}^{\text{env}} \in \mathcal{S}_k.$$

The present manuscript does not claim that.

25.2 Candidate residues

The constructive scan identifies several possible sources of a future suppression witness:

- successor-neighborhood codimension associated with closure stress;
- metric-circuit minimality or compatibility rank;
- deformation-response codimension at the Higgs bridge;
- vacuum-compatibility obstruction index;
- bridge/gravity interface rank;
- residual normal-form index;
- anomaly/gauge-vacuum compatibility residue.

These are candidates because they are finite or obstruction-like. They are not exponents yet.

25.3 Suppression-class non-promotion theorem

Theorem 25.1 (Suppression non-promotion). *No finite residue listed above forces a suppression class for $\rho_{H/G}^{\text{env}}$ unless it is proved to control the closure-scale map σ_{cl} under HEG transport.*

Proof. Suppression forcing requires more than finite structure. It requires a witness-derived exponent or finite class that survives quotienting, transports through the HEG scope, and changes the admissible scale map if altered. A residue that only coexists with the carrier order is not a scale controller. A residue that obtains force through naturalness or measured inputs has hidden selector or calibration ancestry. Therefore finite residue status is insufficient; a control theorem is required. $\square$

25.4 Current suppression endpoint

The current endpoint is

$$\text{Status}_{11} = \text{SuppressionCandidate},$$

not SuppressionForced. This is a constructive status because the paper identifies which kind of theorem would upgrade the endpoint.

26. Relation to charged-lepton and neutrino anchor import

The charged-lepton and neutrino parameter-sector papers show that anchor import is not forbidden in AASC. The rule is order-sensitive:

$$\text{shape or ratio closure first} \Rightarrow \text{one-anchor display second}.$$

For charged leptons, a projective mass-shape object was obtained before one-anchor MeV display. For neutrinos, a splitting-ratio or interval object was admitted before one-anchor spectrum display. In the present manuscript, the analogous closed object is not a singleton ratio but a faithful-projection order class:

$$0 < \rho_{H/G}^{\text{env}} < 1.$$

This permits boundary comparison and bounded one-anchor language, but it does not reconstruct a unique numerical weak scale.

Proposition 26.1 (One-anchor boundary for the HEG witness). *The HEG metric-envelope witness permits one-anchor display only at the level of the closed order class unless a stronger projective or singleton object is later proved.*

Proof. One-anchor import cannot create the quotient relation. It can only calibrate or display an already closed relation. The relation closed here is an inequality, not a projective coordinate or singleton. Therefore one-anchor import may display the bounded relation but cannot determine $\sigma_{\text{cl}}(C_H)$ uniquely. $\square$

27. Relation to standard hierarchy language

The conventional hierarchy problem often begins with a scalar mass term and a high scale. AASC shifts the target from scalar-coordinate smallness to carrier-certified scale comparison. The present witness can be read as answering a prior question:

Before asking why the weak scale has a particular value, what are the standing-bearing carriers whose scale comparison is admissible?

The answer given here is that the admissible denominator is the metric-envelope carrier C_G^{env} , and the admissible numerator is the Higgs bridge/fixation carrier C_H . Their carrier-order relation is forced in the declared HEG scope. Only after that can boundary weak/Planck data, BSM mechanisms, finite residues, or numerical anchor displays be evaluated.

This does not eliminate naturalness or EFT sensitivity. Those remain diagnostics inside physical model scopes. It does prevent them from substituting for the carrier-order witness.

28. Future exactness program

The faithful-projection order result opens three possible exactness paths.

28.1 Suppression-class path

Find a finite residue r such that

$$r \mapsto \mathcal{S}_k$$

and prove that r controls σ_{cl} . Candidate sources include closure-stress codimension, metric-circuit minimality, and bridge/gravity interface rank. This would upgrade

$$0 < \rho_{H/G}^{\text{env}} < 1$$

to

$$\rho_{H/G}^{\text{env}} \in \mathcal{S}_k.$$

28.2 Projective-shape path

Prove a quotient-stable projective class

$$[\sigma_{\text{cl}}(C_H) : \sigma_{\text{cl}}(C_G^{\text{env}})] = [a : b].$$

This would permit one-anchor reconstruction in the charged-lepton style.

28.3 Quotient-singleton path

Prove scale-fiber collapse:

$$\rho_{H/G}^{\text{env}} = \rho_*.$$

This would require obstruction exhaustion, no calibration ancestry, transport closure, and representative independence. Nothing in this manuscript claims that endpoint.

29. Extension checklist for future uses

Any future paper that uses this witness should answer:

1. Is the ratio ρ^{bdry} , ρ^{cand} , or $\rho_{H/G}^{\text{env}}$?
2. Is the denominator C_G^{env} or a Planck label?
3. Is the numerator C_H or a vev-like coordinate?
4. Is σ_{cl} declared?

5. Is σ_{cl} a faithful closure-scale projection of inherited closure load?
6. Are empirical anchors used before or after closure?
7. Is the endpoint carrier order, faithful-projection order type, suppression class, projective class, or singleton?
8. Which alternatives are obstruction-closed?
9. What would falsify the reported endpoint?

30. Conclusion

The no-promotion hierarchy paper showed what does not count as a Higgs/gravity scale derivation. This paper constructs the positive witness needed for the next stage. The denominator is the metric-envelope carrier

$$C_G^{\text{env}} = (L_D, \text{MetricCircuit}_D, \text{BoundTrans}_D, \text{ETC}_D),$$

not a Planck label. The numerator is the Higgs bridge/fixation carrier

$$C_H = H_{\text{br/fix}},$$

not a vev, pole mass, potential coefficient, RG coordinate, or fitted weak scale. Standing monotonicity and the RPI role ladder show that the Higgs bridge carrier is carrier-subordinate to the metric-envelope carrier:

$$C_H \triangleleft C_G^{\text{env}}.$$

Inherited closure load, successor-neighborhood thinness, and projection faithfulness then supply the discipline required to turn carrier order into a closure-scale hierarchy. For any faithful closure-scale projection σ_{cl} , the close witness yields

$$0 < \rho_{H/G}^{\text{env}} = \frac{\sigma_{\text{cl}}(C_H)}{\sigma_{\text{cl}}(C_G^{\text{env}})} < 1.$$

This is a genuine positive hierarchy result, but it is not the observed electroweak/Planck ratio. It is the faithful-projection proper order type for the metric-envelope/Higgs-bridge closure-scale ratio. The explicit numerical ratio, one-anchor reconstruction to an empirical scale, and finite suppression-class forcing remain open until additional identification or finite-residue control theorems are supplied.

In compressed form:

$$\text{standing monotonicity} + \text{metric-envelope closure} + \text{RPI/Higgs bridge} \Rightarrow C_H \triangleleft C_G^{\text{env}} \Rightarrow 0 < \rho_{H/G}^{\text{env}} < 1.$$

The explicit ratio is not obtained by importing anchors. It is obtained, if at all, by first closing the Higgs/gravity scale fiber. This paper closes the projected faithful-projection proper order type of that fiber for faithful closure-scale projection and states the remaining obstruction honestly.

A. Corpus and source dependency ledger

Source/module	Imported result	Role in this paper
Arrow of Time as Standing Monotonicity	Standing precedence on raw stages; quotient partial order after standing equivalence; no-backflow discipline	carrier-order machinery
Early Metric-Interior Transport Closure	first licensed metric-comparison regime L_D , metric circuits, closure stress, early transport/closure certificate	denominator carrier and closure-scale order
Realized Physical Interior	role ladder through comparison, transport, field, particle, gauge, mass, vacuum; Higgs bridge/fixation carrier in SM tuple	numerator carrier and dependency order
ZPE trilogy	renormalization quotient as standing preservation; scalar offset non-promotion	scalar non-promotion precedent
No-promotion hierarchy paper	HEG redescription ledger, boundary-ratio separation, Planck-label quotient, naturalness promotion boundary	false-anchor firewall
CNF8 hierarchy/ratio forcing	hierarchy type and suppression class status criteria	endpoint classification
CS4/CS5 scale-fiber certificates	scale-fiber declaration, one-anchor display rules, no amplification	certificate templates
Charged-lepton ratio arc	shape before one-anchor rendering	lawful anchor analogy
Neutrino ratio arc	standing ratio/interval before absolute mass	lawful ratio analogy

B. Certificate checklist

1. Is the scope S_{HG} fixed?
2. Is C_G^{env} defined without Planck-label substitution?
3. Is C_H defined without vev or pole-mass substitution?
4. Is standing precedence stated in quotient form?
5. Are equality, reverse order, and incomparability obstruction-closed?
6. Is closure stress defined by successor-neighborhood inclusion?
7. Is σ_{cl} explicitly declared as a faithful closure-scale projection?
8. Is calibration ancestry excluded?
9. Is the endpoint limited to the strongest closed status?
10. Are boundary display, faithful-projection order type, magnitude ratio, suppression class, and singleton ratio distinguished?

C. Formalization boundary

This specific witness-construction paper has not yet received a paper-specific Lean 4 formalization. The general AASC formalism is deeply mechanized elsewhere in the corpus. A future formalization of this paper would target the following structures: carrier admission, standing precedence, carrier subordination, successor-neighborhood closure stress, closure-scale projection, endpoint classification, and one-anchor reconstruction boundary. It would not attempt to formalize numerical Standard Model phenomenology in this paper.

D. Notation index

Symbol	Meaning
S_{HG}	fixed Higgs/gravity comparison scope
C_G^{env}	metric-envelope carrier: L_D , metric circuit, bounded transport, and ETC
C_H	Higgs bridge/fixation carrier $H_{\text{br/fix}}$
L_D	first licensed metric-comparison regime
MetricCircuit_D	standing-preserving redescription loop for metric comparison content
BoundTrans_D	bounded perturbation transport discipline
ETC_D	early transport/closure certificate
$\text{Succ}_C(x)$	standing-preserving admissible successor neighborhood of x
$\text{ClStress}_C(x)$	closure-stress order defined by successor-neighborhood thinness
$\preceq_S$	standing precedence
$\triangleleft$	carrier subordination
σ_{cl}	closure-scale projection
$\rho_{H/G}^{\text{env}}$	metric-envelope/Higgs-bridge closure-scale ratio
$\rho_{H/G}^{\text{bdry}}$	empirical boundary display $v_F/\bar{M}_{\text{Pl}}$
Endpoint_{HG}	strongest status reported by the witness

E. Extended source import notes

E.1 Arrow paper import

The Arrow paper is imported only for standing monotonicity and no-backflow. It is not imported for a physical entropy result, cosmological initial condition, or thermodynamic mechanism. Its relevance here is the quotient-order form: standing precedence on raw role stages descends to a partial order on standing-equivalence classes.

E.2 Early metric-interior import

The Early Metric paper is imported for the first licensed metric-comparison regime, metric circuits, closure stress, and early transport/closure certificates. Its inflationary interpretation is not used to derive the hierarchy result. The current paper uses the structural metric-envelope carrier, not a scalar-field inflation model.

E.3 RPI import

The RPI paper is imported for the role ladder and the Higgs bridge/fixation carrier. The Standard Model structural specialization is used to type $H_{\text{br/fix}}$, not to import Higgs-sector numerical values.

F. Publication-facing theorem ladder

1. Metric comparison is licensed only through standing-preserving closure.
2. The metric-envelope carrier is built from first metric comparison, metric circuits, bounded transport, and early transport/closure.
3. Mass/deformation response is downstream of comparison, transport, fields, particles, and gauge.
4. The Higgs bridge/fixation carrier is a proper interior response carrier, not a scale value.
5. The metric-envelope carrier is standing-precedent to the Higgs bridge carrier.
6. Equality, reverse order, and incomparability are obstruction-closed.
7. Faithful closure-scale projection reports strict inherited closure-load order as strict positive projected order.
8. Therefore $0 < \rho_{H/G}^{\text{env}} < 1$ as a faithful-projection order display, while magnitude remains open by projection non-uniqueness.
9. Suppression and singleton endpoints require additional finite-residue or scale-fiber collapse theorems.

G. Extension protocol

Any extension attempting to derive a more explicit ratio should declare:

1. whether it targets $\rho_{H/G}^{\text{env}}$, $\rho_{H/G}^{\text{bdry}}$, or a new physical ratio;
2. the numerator carrier and whether it is structural, boundary-normalized, or empirical;
3. the denominator carrier and whether it is structural, boundary-normalized, or empirical;
4. the scale projection and its monotonicity or collapse rule;
5. the calibration ancestry graph;
6. the finite residue or projective object, if any;
7. the status of one-anchor import;
8. the exact endpoint: boundary display, faithful-projection order type, magnitude ratio, suppression class, projective class, or singleton.

References

- [1] Amos Jay Maley. *The Arrow of Time as Standing Monotonicity: Directionality, Records, Entropy, Causality, and Cosmological Asymmetry under AASC*. 2026.
- [2] Amos Jay Maley. *Early Metric-Interior Transport Closure: The Structural Role of Inflationary Explanation under AASC*. 2026.
- [3] Amos Jay Maley. *The Realized Physical Interior: Locality, Metric Role, Fields, Particles, Gauge Structure, Mass, Vacuum, Time, Causality, and the Standard Physics Stack under AASC*. 2026.
- [4] Amos Jay Maley. *The Hierarchy Problem and Scale Standing: An AASC No-Promotion Theorem for Higgs/Electroweak/Gravity Comparisons*. 2026.
- [5] Amos Jay Maley. *Why Zero-Point Vacuum Energy Is Not a Cosmological Prediction in Quantum Field Theory*. Zenodo, 2026. doi: 10.5281/zenodo.20042236.
- [6] Amos Jay Maley. *Invariance Constraints and the Non-Predictivity of Zero-Point Vacuum Energy in Quantum Field Theory*. Zenodo, 2026. doi: 10.5281/zenodo.20044293.
- [7] Amos Jay Maley. *Renormalization Invariance and Standing Preservation in the Non-Predictivity of Zero-Point Vacuum Energy*. Zenodo, 2026. doi: 10.5281/zenodo.20044419.
- [8] Leonard Susskind. Dynamics of spontaneous symmetry breaking in the Weinberg-Salam theory. *Physical Review D* 20, 2619–2625, 1979.
- [9] Gerard 't Hooft. Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking. In *Recent Developments in Gauge Theories*, NATO Advanced Study Institutes Series, 1980.
- [10] Martinus J. G. Veltman. The infrared-ultraviolet connection. *Acta Physica Polonica B* 12, 437, 1981.
- [11] Eldad Gildener. Gauge-symmetry hierarchies. *Physical Review D* 14, 1667–1672, 1976.
- [12] Kenneth G. Wilson and John Kogut. The renormalization group and the epsilon expansion. *Physics Reports* 12, 75–199, 1974.
- [13] Joseph Polchinski. Renormalization and effective Lagrangians. *Nuclear Physics B* 231, 269–295, 1984.
- [14] Gian F. Giudice. Naturally speaking: the naturalness criterion and physics at the LHC. In *Perspectives on LHC Physics*, World Scientific, 2008.
- [15] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.
- [16] Steven Weinberg. *The Quantum Theory of Fields, Volume I*. Cambridge University Press, 1995.